

RESEARCH INTERESTS

My research interests lie in developing **AI-powered medical solutions**, with a focus on mathematically grounded modeling and foundation-model-driven designs. I have extensive experience in **learning with imperfect biomedical data**, particularly in semi/unsupervised learning, transfer learning, and deep multimodal learning under domain discrepancy and low-quality labeling scenarios.

EDUCATION

Ho Chi Minh City University of Technology

B.E. in Computer Engineering (High-Quality Program)

- **GPA:** 9.27/10.0 (3.9/4.0) [[Transcript](#)]

- **Valedictorian - Ranked 1st** out of 749 students in the Department of Computer Science and Engineering.

Sep 2021 – Nov 2025

HCMC, Vietnam

PROFESSIONAL EXPERIENCE

Xu Lab - Carnegie Mellon University [🔗](#)

Research Assistant

- Advisors: [Prof. Min Xu](#), [Dr. Xingjian Li](#) and [Prof. Tianyang Wang](#)

- Integrated the Mixture-of-Experts concept to address the homogeneity problem in semi-supervised learning in the "Semi-MoE" project.

- Led the "SemiSAM" project, developing a bidirectional co-training framework that aligned foundation and conventional models for effective utilization of unlabeled data.

- Assisted in the project "Bridging multiple domain shifts in mixed-domain medical image segmentation".

- Assisted in drafting peer-review feedback for conference papers (e.g., IJCAI, MICCAI).

Aug 2024 - Present

Pennsylvania, USA

Machine and Hybrid Intelligence Lab - Northwestern University [🔗](#)

Research Assistant

- Advisor: [Prof. Ulas Bagci](#)

- Developed a mean-teacher framework to disentangle stain appearance and tissue structure of the histopathology data under significant H&E variability and annotation scarcity.

- Assisted in the projects "Source-Free Domain Adaptation in Medical Image Segmentation" and "Radiomics Analysis of Laryngeal Cancer."

Jun 2025 - Present

Illinois, USA

AIVIETNAM [🔗](#)

Research Associate, Teaching Assistant

- Advisors: [Dr. Quang Vinh Dinh](#)

- Co-Lead of AIMA: AI-in-Medicine Applications lab, leading 4 teams in Medical AI.

- Providing academic support for 600+ students in the All-in-One 2024 course and hosting online seminars on semi-supervised learning and knowledge distillation.

May 2024 - Present

HCMC, Vietnam

Integrated MechanoBioSystems Lab - National Cheng Kung University [🔗](#)

AI Research Intern

- Advisor: [Ting-Yuan Tu](#)

- Developed a Y-Net Siamese architecture for fast 3D single-cell tracking and segmentation in 3D time-lapse images.

Jul 2025 - Sep 2025

Tainan, Taiwan

FPT Software AI Center - FPT Corporation [🔗](#)

AI Researcher

- Advisor: [Prof. Anh Nguyen](#)

- Applied denoising diffusion implicit models (DDIM) for interactive 3D editing in "DragPC - point cloud shape deformation via latent point diffusion models" project.

- Hosted weekly seminar sessions on diffusion, NeRF, 3D Gaussian splatting, and large vision models.

Jun 2024 - Sep 2024

Hanoi, Vietnam

PUBLICATIONS

C=CONFERENCE, J=JOURNAL

[C.11] Semi-MoE: Mixture-of-Experts meets Semi-Supervised Histopathology Segmentation

Vi Vu*, TH Nguyen*, HT Nguyen, D Kihara, T Wang, X Li, M Xu

BMVC - British Machine Vision Conference (2025). [[PDF](#)].

[C.10] From Specialist to Generalist: Unlocking SAM's Learning Potential on Unlabeled Medical Images

Vi Vu*, HT Nguyen*, TT Nguyen*, TB Lam, TH Nguyen, T Wang, X Li, M Xu

ISBI - IEEE 23rd International Symposium on Biomedical Imaging (2026). [[PDF](#)].

- [C.9] **Semi-Supervised Histopathology Image Segmentation with Feature Diversified Collaborative Learning**
Vi Vu*, TH Nguyen*, HT Nguyen, QV Dinh, X Li, M Xu
AAAI Bridge Program - AI for Medicine and Healthcare Bridge program (2025). [\[PDF\]](#).
- [C.8] **HDC: Hierarchical Distillation for Multi-level Noisy Consistency in Semi-Supervised Fetal Ultrasound Segmentation**
Vi Vu*, TQK Le*, HH Pham, XL Huynh, TH Nguyen, MHN Le, Q Nguyen, HD Nguyen
CVPRW - The 7th IEEE/CVF CVPR Precognition Workshop (2025). [\[PDF\]](#).
- [C.7] **Learning Disentangled Stain and Structural Representations for Semi-Supervised Histopathology Segmentation**
Vi Vu*, HH Pham*, TH Nguyen, U Bagci, M Xu, TN Le, HH Pham
COMPAYL - MICCAI Workshop on Computational Pathology with Multimodal Data (2025). [\[PDF\]](#).
- [C.6] **Semi-Supervised Skin Lesion Segmentation under Dual Mask Ensemble with Feature Discrepancy Co-Training**
TH Nguyen, T Nguyen, XB Nguyen, Vi Vu, VQ Dinh, F MERIAUDEAU
MIDL - Medical Imaging with Deep Learning (2025). [\[PDF\]](#).
- [C.5] **Fetal-BCP: Addressing Empirical Distribution Gap in Semi-Supervised Fetal Ultrasound Segmentation**
HH Pham, HT Nguyen, Vi Vu, QV Dinh, TH Nguyen, X Li, M Xu
ISBI - IEEE 22nd International Symposium on Biomedical Imaging (2025). [\[PDF\]](#).
- [C.4] **Domain-Invariant Mixed-Domain Semi-Supervised Medical Image Segmentation with Clustered Maximum Mean Discrepancy Alignment**
TH Nguyen, BT Lam, Thien Nguyen, TQK Bui, Vi Vu, KP Huynh, U Bagci, M Xu
ICASSP - International Conference on Acoustics, Speech, and Signal Processing (2026). [\[PDF\]](#).
- [C.3] **Aligning What You Separate: Denoised Patch Mixing for Source-Free Domain Adaptation in Medical Image Segmentation**
T Nguyen, TH Nguyen, TQK Bui, BT Lam, Vi Vu, KP Huynh, , U Bagci, M Xu
ICASSP - International Conference on Acoustics, Speech, and Signal Processing (2026). [\[PDF\]](#).
- [C.2] **Hepatic Tumor Segmentation under Modified Scalable and Transferable nnU-Net Framework**
TQK Bui, M. Dinh, Vi Vu, Q. Nguyen, Q. Dinh, M. Le, Q. Le
CITA - Conference on Information Technology and its Applications (2025). [\[PDF\]](#).
- [C.1] **Modality-Specific Enhancement and Complementary Fusion for Semi-Supervised Multi-Modal Brain Tumor Segmentation**
DT Chung, BT Lam, TH Nguyen, T Nguyen, Vi Vu, HL Cao, PK Huynh, M Xu
AIMedHealth26 - AI for Medicine and Healthcare Bridge Program at AAAI26 (2026). [\[PDF\]](#).
- [J.4] **Adaptive Knowledge Transferring with Switching Dual-Student Framework for Semi-Supervised Medical Image Segmentation**
TH Nguyen, HT Nguyen, Vi Vu, BT Lam, BX Nguyen, J Xing, T Wang, X Li, M Xu
Pattern Recognition (IF 7.6) (2025). [\[PDF\]](#).
- [J.3] **DuetMatch: Harmonizing Semi-Supervised Brain MRI Segmentation via Decoupled Branch Optimization**
TH Nguyen, HT Nguyen, Vi Vu, BT Lam, P Huynh, T Wang, X Li, U Bagci, M Xu
Computerized Medical Imaging and Graphics (IF 4.9) (2025). [\[PDF\]](#).
- [J.2] **UP2D: Uncertainty-aware Progressive Pseudo-label Denoising for Source-Free Domain Adaptive Medical Image Segmentation**
TQK Bui, T. Nguyen, D. Ho, B. Lam, Vi Vu, T. Nguyen, P. Huynh, U. Bagci
Neurocomputing (IF 6.5) (2026). [\[PDF\]](#).
- [J.1] **Performance analysis of artificial intelligence-driven convolutional neural network architectures for liver tumor segmentation**
MHN Le, TH Nguyen, MT Dinh, TQK Bui, Vi Vu, HQ Kha, PK Nguyen, NHH Le, TM Nguyen, HH Huynh, K Le
Journal of Clinical Oncology (IF 42.1) (2025). [\[PDF\]](#).

HONORS AND AWARDS

Winner of the Fetal Ultrasound Grand Challenge at ISBI in Texas, USA [\[Cert\]](#)
 NSTC Scholarship for International Internship Pilot Program in Tainan, Taiwan [\[Cert\]](#)
 Achieve outstanding Academic Performance certification in HCMC, Vietnam [\[Cert\]](#)
 1st Prize, City-level Excellent Physics Student in HCMC, Vietnam [\[Cert\]](#)
 Momentum Scholarship (top 10 university students) in HCMC, Vietnam
 4 Academic Incentive Scholarships Type 1 (top 1.6% of university students) in HCMC, Vietnam
 2 OISP Scholarships Type 1 for high-quality English-taught programs in HCMC, Vietnam
 Achieve the title of Outstanding All-around Student in HCMC, Vietnam